

Forecasting Research on Long-term Solar Irradiance with An Improved Prophet Algorithm

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Abstract: To solve the problem of predicting solar irradiance in integrated energy systems (IES), this paper improves the Prophet algorithm by removing the trend and holidays terms, changing to the multiplicative form, and adding the monthly regressors. The improved Prophet algorithm is compared with the original and autoregressive integrated moving average (ARIMA) algorithms respectively, and the forecast accuracy is significantly improved, and the RMSE of the long-term forecast is 108.30% lower than that of ARIMA. The improved algorithm can be applied to forecast solar irradiance on long time scales with single information, providing a solution to the problem of PV array planning for IES. At the same time, the forecast data obtained by the algorithm can also be used as a reference value to optimize the day-ahead dispatch.

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1. INTRODUCTION

Nowadays, Integrated energy system(IES) contains multiply energy types, besides conventional fossil fuels like coal, gas and so on, it also includes renewable energy like solar power and wind power. Renewable energy has a more and more essential position with the greenhouse effect and air pollution and takes an important part in Europe's energy combination (Renewable energy statistics, 2018). But a problem comes with the stochasticity of renewable energy, which increases the intractability in planning, dispatching, and operating. In renewable energy, solar energy takes an important part. The forecasting research on solar irradiance is necessary as it calculates the potential of an area before planning and provides advice on control after running.

The forecast is multiple time scales and every scale has unique requirements. Short-term forecast fulfills control and dispatch, which has many methods. These methods are divided into two parts: the physical model and the mathematical model. The physical model forecasts solar irradiance by analyzing the physical process in radiation transfer, which is like the Physics-based Smart Persistence model used by Kumler et al., 2019. The mathematical methods are listed: Support Vector Machine (SVM) (Jiang et al. 2017), autoregressive integrated moving average (ARIMA) (Reikard., 2009. Cheng et al., 2012), long short-term memory (LSTM) networks (Qing et al., 2018) et al, which forecasting the irradiance using a fitting model based on the former data set. There also are some researches combining the physical and mathematical methods, like Ping et al. (2019) and Srivastava et al. (2018). But Almost all researches proposed to focus on short-term solar irradiance forecasting, the researches on long-term forecast are far fewer than short-term's. Different from the short-term forecast, the

long-term forecast is meaningful in the address choosing of the solar field. In the design of the solar field, the long-term solar irradiance should be taken into account arising from the optimal economic effectiveness. Verbois et al. (2018) implement WRF in three years of hourly irradiance forecast with the result that the Root Mean Square Error (RMSE) 23% is lower than smart persistence which still is 169. The stochastic accumulates with the forecast time increasing, which also accumulates more and more bias. It is a hint that solar irradiance is a time series with strong seasonality.

The Prophet time series algorithm was proposed by Facebook in 2017 (Taylor et al., 2017). Its main application scenario is time series forecasting. Since the Prophet algorithm still has its shortcomings, it cannot be directly applied to solar irradiance forecasting. In this paper, the Prophet algorithm should be firstly improved, then it tries to predict solar irradiance using the improved Prophet algorithm, and finally compares the forecasting results with those of the ARIMA algorithm to conclude.

2. IMPROVED PROPHET ALGORITHM

2.1 Introduction of Prophet Algorithm

Prophet divides the predicted time series into four parts: trend, seasonality, holiday, and errors, which is different from other traditional algorithms. Compared with other traditional time series forecasting algorithms, Prophet has advantages in fast-fitting, strong scalability, and simple parameter adjustment, and are suitable for non-professionals in time series forecasting. Its main components are shown in equation

$$y(t) = g(t) + s(t) + h(t) + \varepsilon_t \quad (1)$$

where $g(t)$ is the trend effect, represents the trend of time series; $s(t)$ is seasonality effect, represents the seasonality of time series; $h(t)$ is the holiday effect, represents that the day is a holiday or not; ε_t is an error, represents the possible deviation of the forecasting value.

2.2 Improved of Prophet Algorithm

2.2.1 Remove the trend and holidays

According to the paper proposed, there are two kinds of trend effect, saturating growth model and piecewise linear model. However, the introduction of trend terms presupposes that the data itself is trending, and there is no strong trend in solar irradiance. According to the literature, the trend in solar irradiance for some frequency bands in North America after 1990 is not significant (Fioletov et al., 2001), and the trend term causes large deviations in the forecast of solar irradiance with increasing forecasting time, as experimentally illustrated in Chapter 3. In the same way, the solar irradiance is low relative to the holiday effect which also can be removed.

So the Prophet equation without trend term is proposed:

$$y(t) = C + s(t) + \varepsilon_t \quad (2)$$

where C is the foundational solar irradiance which is invariant by time.

It is important to note that this paper does not deal with the calculation of a particular value that has already been proposed in the preceding paper.

2.2.2 Changing to the multiplicative form

The original Prophet algorithm is only applicable to data with small periodic amplitudes. However, for solar irradiance, which all is zero in the night. So if the components in the forecasting equation are in additive form, all situations will not be satisfied. Therefore, if the components are multiplied, the daily seasonality ratios are set to 0 for a night, which is the better approach to forecast solar irradiance.

The equation is below:

$$y(t) = [1 + \hat{s}(t)] \cdot C + \varepsilon_t \quad (3)$$

$$\hat{s}(t) \in [-1, +\infty)$$

where $\hat{s}(t)$ is a transformed seasonality in percentage.

2.2.3 Add monthly regressors

Solving the problem of which solar irradiance is 0 at night is not enough. It is more about making accurate forecasting of solar irradiance during the day. One of the biggest difficulties is that there is a large difference between the daily peak in winter and summer. To solve this problem, the algorithm

involves monthly regressors. It is a conditional seasonality, which adds up the equation when the time reaches a preset condition. In this way, an appropriate correction can be done and solve the problem of significant differences in amplitude changes between winter and summer.

So the equation is:

$$y(t) = [r(t) + C] \cdot [1 + \hat{s}(t)] + \varepsilon_t \quad (4)$$

s.t. C is constant

where $r(t)$ is the monthly regressors, which forecast the solar irradiance in a month.

Note $R(t) = [r_{Jan}(t) \ r_{Feb}(t) \ \dots \ r_{Dec}(t)]_{1 \times 12}$ and $E = [isJan \ isFeb \ \dots \ isDec]^T_{12 \times 1}$, so $R(t)$ is a vector that consists of monthly regressors, and E is a Boolean vector to justify the month. The equation can be transformed into:

$$y(t) = [R(t)E + C] \cdot [1 + \hat{s}(t)] + \varepsilon_t \quad (5)$$

All the parameters in this equation should be determined by the Limited-memory Broyden–Fletcher–Goldfarb–Shanno algorithm (L-BFGS) to find a maximum posterior estimate, which can solve the optimizing problem with limited memory (Carpenter et al., 2017). The optimization algorithm is not the key point in this paper, so the math derivation is not put forward.

3. RESULTS AND ANALYSIS OF SIMULATION

3.1 Comparison Improve Prophet algorithm with its origin

Both algorithms are implemented in the solar irradiance forecasting in the central region of the U.S (NOAA solar irradiance data). The training data is from 1998–2003, the test data is from 2004–2005. All the data have been pre-processed to remove all points less than zero. Select two typical dates for winter and summer in 2005, which presents the comparison of forecasting results between two different algorithms.

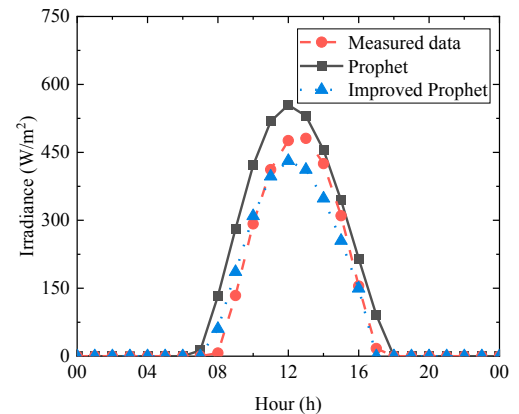


Fig.1. Forecasting and actual solar irradiance on January 1, 2005.

As can be seen from Fig.1, Fig.2 and Fig.3, the forecasting of solar irradiance by the original Prophet algorithm is

unsatisfactory, and the deviations between the fitted values and the actual values are large, mainly due to the following three points:

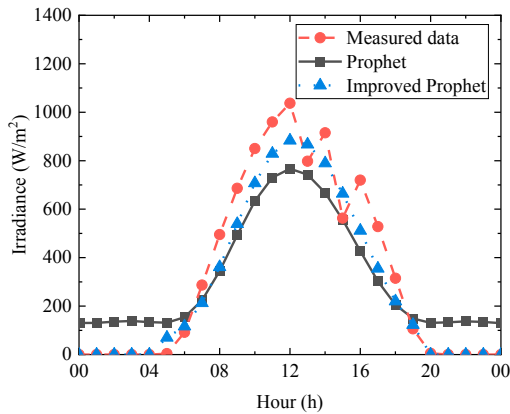


Fig.2. Forecasting and actual solar irradiance on July 1, 2005.

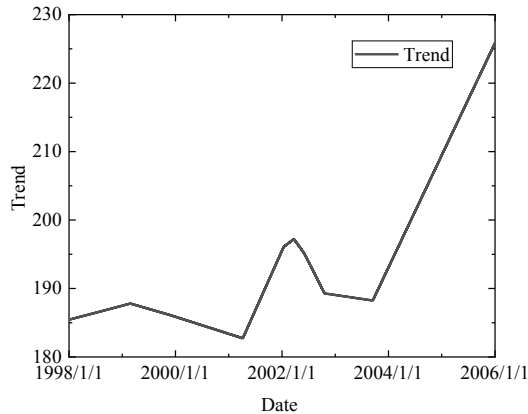


Fig.3. Trend effect in original Prophet algorithm

- (1) Summer forecasting deviates greatly from the actual value. Although the daily trend is correct, the summer data deviates greatly. Take July 1 as an example, the difference of light intensity measurement, that is, the difference between daily solar irradiance maximum and daily solar irradiance minimum, deviated greatly from the actual value. On July 1, The predicted value is 636 W/m² and the actual value is 1037W/m², a deviation of 38.67%; on January 1, the deviation was 10.40%, a significant decrease compared to the summer.
- (2) The forecasting values are not consistent with common sense. Figure 2 shows that the forecasting solar irradiance is still not zero at night in summer, which is not consistent with actual common sense. Also note that in winter and summer, the predicted amplitudes are 633.92 W/m² and 636.49 W/m², respectively, which are not very different.
- (3) The trend is wrong. The forecasting is going up with an incorrect trend, which deviates greatly from the actual, and is more obvious in the figure. By December 31, 2005, the maximum solar irradiance changed to 530 W/m², which deviated from the actual value of 457

W/m² by 15.97%, which increases from the previous year.

These problems are caused by two main factors:

- (1) Prophet's algorithm is composed of four parts, in which the seasonality is obtained by Fourier transform and then added to the original trend, and it is suitable for data with little deviation in amplitude at each change period, so there will be similar amplitudes of winter and summer solar irradiance forecasting. However, for solar irradiance, the deviation of solar irradiance between winter and summer is large, and the daily seasonality cannot be added to cover the original characteristics, so there is a large deviation in the summer data.
- (2) Prophet algorithm contains a trend effect, which is fitted according to the training data, and the resulting function is segmented; beyond the range of training data, the last segmented function will be used to forecast the future value, and the forecasting value is the value of the segmented function plus seasonality. Since the slope is constant, the absolute value of the segmentation function will always increase, and the forecasting error will naturally increase with time.

The RMSE is 204.31 W/m² for the ordinary Prophet algorithm and 135.47 W/m² for the improved Prophet algorithm in annual forecasting, which is a significant improvement. It is also obvious that the improved Prophet algorithm circumvents these problems wonderfully. So, it proves that the improved Prophet algorithm is feasible in solar irradiance forecasting.

3.2 Comparison Improve Prophet algorithm with its ARIMA

Compare Improved Prophet algorithm with ARIMA with the same data in 3.1, and ARIMA uses the auto_arima function in Python module named pmdarima. This figure shows a comparison between each other.

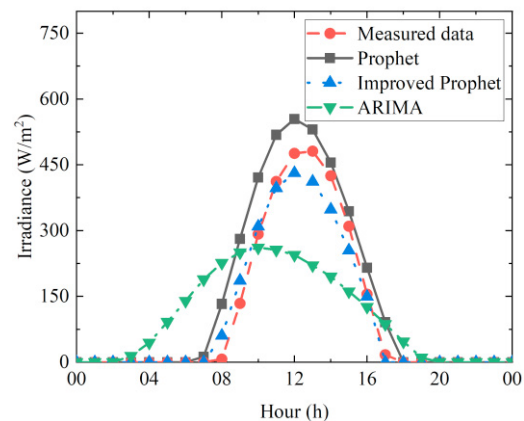


Fig.4. Forecasting and actual solar irradiance on January 1, 2005.

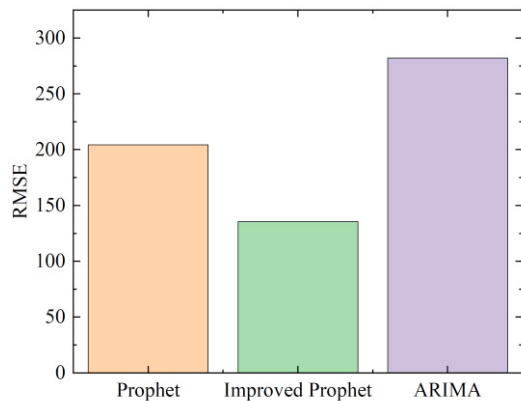


Fig.5. RMSE comparison for three forecasting algorithms

From Fig.4, the ARIMA algorithm has worse short-term forecasting accuracy than the improved Prophet algorithm. And for long-term forecasting, the RMSE of ARIMA is 282.18 which is 108.30% larger than the Improved Prophet algorithm. It shows obviously in Fig.5

4. CONCLUSIONS

In this paper, the original Prophet algorithm is improved by removing the trend and holidays, changing to the multiplicative form, and adding monthly regressors. The improved Prophet algorithm proposed in this paper has significantly improved the solar irradiance forecasting compared with the Prophet algorithm. Compared with the commonly used ARIMA algorithm, the short-term prediction accuracy of this algorithm is similar, and the long-term prediction accuracy is 108.30% higher than ARIMA. Therefore, the algorithm can apply a single-information long-term solar irradiance forecasting for planning the siting of PV arrays in IES. Later research can introduce uncertainty to this deterministic prediction algorithm to achieve probabilistic predictions, or add regressors to expand it to include long-term predictions with multiple information quantities, such as temperature and humidity.

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